

Dark States and Coherent Population Trapping

Quantum Interference, Λ -Systems, and Applications in Quantum Control

Introduction and Physical Motivation

When a quantum system is driven by two coherent fields, destructive quantum interference can prevent any absorption of light, even when the atom is in a state that naively couples to both fields. This phenomenon — **Coherent Population Trapping (CPT)** — results in the population becoming “trapped” in a **dark state**: a superposition of ground states that is completely decoupled from the excited state by quantum interference.

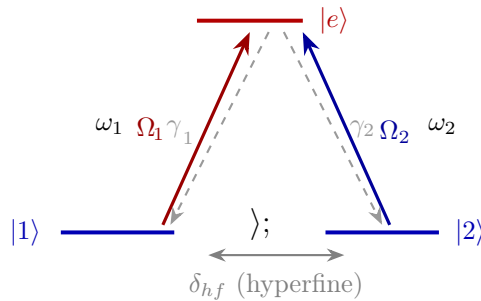
CPT was first observed experimentally by Alzetta *et al.* (1976) as a sudden disappearance of fluorescence (a “dark line”) when two laser frequencies satisfied a two-photon resonance condition. The explanation in terms of dark states was given by Arimondo and Orriols, and developed extensively by Aspect, Arimondo, and others.

Why this matters: Dark states are at the heart of laser cooling below the recoil limit (velocity-selective CPT), electromagnetically induced transparency (EIT), slow light, quantum memory, stimulated Raman adiabatic passage (STIRAP), and qubit state preparation. Understanding dark states is thus essential across laser physics, atomic physics, and quantum information.

The Λ -System: Three-Level Atom

Level Structure

The simplest system exhibiting CPT is the **Λ -system** (named for its shape in an energy diagram):



The three states are:

- $|1\rangle, |2\rangle$: two **ground states** (e.g., hyperfine sublevels), connected by a forbidden or very weakly allowed transition (no direct dipole coupling between them)

- $|e\rangle$: an **excited state** that decays back to $|1\rangle$ and $|2\rangle$ with rates γ_1 and γ_2
- Ω_1, Ω_2 : Rabi frequencies of two driving laser fields coupling $|1\rangle \leftrightarrow |e\rangle$ and $|2\rangle \leftrightarrow |e\rangle$

The key structural feature is that **there is no direct coupling between $|1\rangle$ and $|2\rangle$** ; both connect only to $|e\rangle$. This creates two interfering pathways to the excited state.

The Dark State: Derivation from First Principles

Interaction Picture Hamiltonian

Working in the rotating frame (interaction picture) with the drive frequencies, the Λ -system Hamiltonian in the basis $\{|e\rangle, |1\rangle, |2\rangle\}$ is:

$$H = \hbar \begin{pmatrix} 0 & \Omega_1^*/2 & \Omega_2^*/2 \\ \Omega_1/2 & -\delta_1 & 0 \\ \Omega_2/2 & 0 & -\delta_2 \end{pmatrix} \quad (1)$$

where $\delta_1 = \omega_1 - \omega_{e1}$ and $\delta_2 = \omega_2 - \omega_{e2}$ are the single-photon detunings from resonance.

The Two-Photon Resonance Condition

The crucial condition for CPT is the **two-photon resonance**:

$$\delta_1 = \delta_2 \equiv \delta \quad \Longleftrightarrow \quad \omega_1 - \omega_2 = \omega_{e1} - \omega_{e2} \quad (2)$$

This means the *difference* of the drive frequencies exactly matches the ground-state splitting, even if neither drive is individually resonant with the excited state.

Identifying the Dark State

We look for a superposition of the ground states:

$$|D\rangle = c_1 |1\rangle + c_2 |2\rangle \quad (3)$$

that is **completely decoupled from $|e\rangle$** , i.e.,

$$\langle e | H | D \rangle = 0 \quad (4)$$

Evaluating using Eq. (1):

$$\langle e | H | D \rangle = \frac{\hbar}{2} (\Omega_1^* c_1 + \Omega_2^* c_2) = 0 \quad (5)$$

This gives $c_1/c_2 = -\Omega_2^*/\Omega_1^*$. After normalisation:

$$|D\rangle = \frac{\Omega_2 |1\rangle - \Omega_1 |2\rangle}{\sqrt{|\Omega_1|^2 + |\Omega_2|^2}} \quad (6)$$

The dark state is the **null space** of the coupling operator $V = \Omega_1 |e\rangle\langle 1| + \Omega_2 |e\rangle\langle 2|$. It exists regardless of the specific values of Ω_1 and Ω_2 — as long as both drives are non-zero and the two-photon resonance condition Eq. (2) holds. No light is absorbed from $|D\rangle$ because the two excitation pathways $|1\rangle \rightarrow |e\rangle$ and $|2\rangle \rightarrow |e\rangle$ interfere *destructively*.

The Bright State

Orthogonal to $|D\rangle$ is the **bright state**:

$$|B\rangle = \frac{\Omega_1^* |1\rangle + \Omega_2^* |2\rangle}{\sqrt{|\Omega_1|^2 + |\Omega_2|^2}} \quad (7)$$

which couples *maximally* to the excited state:

$$\langle e|H|B\rangle = \frac{\hbar}{2} \sqrt{|\Omega_1|^2 + |\Omega_2|^2} \equiv \frac{\hbar\Omega_{\text{eff}}}{2} \quad (8)$$

In the $\{|D\rangle, |B\rangle, |e\rangle\}$ basis, the Hamiltonian becomes block-diagonal: $|D\rangle$ is completely decoupled, while $|B\rangle$ and $|e\rangle$ form a simple two-level system driven at the effective Rabi frequency $\Omega_{\text{eff}} = \sqrt{|\Omega_1|^2 + |\Omega_2|^2}$.

Density Matrix and Optical Bloch Equations

The Lindblad Master Equation

To include spontaneous emission, we use the Lindblad master equation:

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \gamma_1 \left(\sigma_{1e} \rho \sigma_{1e}^\dagger - \frac{1}{2} \{ \sigma_{1e}^\dagger \sigma_{1e}, \rho \} \right) + \gamma_2 \left(\sigma_{2e} \rho \sigma_{2e}^\dagger - \frac{1}{2} \{ \sigma_{2e}^\dagger \sigma_{2e}, \rho \} \right) \quad (9)$$

where $\sigma_{ie} = |i\rangle\langle e|$ is the decay operator from $|e\rangle$ to $|i\rangle$, and γ_i is the spontaneous emission rate into channel i .

Steady State: Population Trapping

At steady state ($\dot{\rho} = 0$), if the two-photon resonance condition holds, the solution is:

$$\rho_{ee}^{\text{ss}} = 0, \quad \rho_{DD}^{\text{ss}} = 1 \quad (10)$$

All population accumulates in the dark state $|D\rangle$. The excited state is completely depopulated — hence zero fluorescence. This is Coherent Population Trapping.

The Off-Resonance Case

Away from two-photon resonance (Raman detuning $\delta_R = \delta_1 - \delta_2 \neq 0$), the dark state is no longer stationary. The population oscillates with a Raman frequency δ_R and fluorescence is restored. The CPT resonance has a linewidth:

$$\Gamma_{\text{CPT}} \approx \frac{|\Omega_1|^2 + |\Omega_2|^2}{\gamma} \quad (11)$$

which can be *far narrower* than the natural linewidth γ when the drives are weak ($|\Omega_{1,2}| \ll \gamma$). This is the basis for ultra-narrow spectral features used in atomic clocks.

Dressed-State Picture and Adiabatic Elimination

The Dressed Basis Transformation

Defining the mixing angle θ by:

$$\cos \theta = \frac{|\Omega_1|}{\Omega_{\text{eff}}}, \quad \sin \theta = \frac{|\Omega_2|}{\Omega_{\text{eff}}}, \quad \Omega_{\text{eff}} = \sqrt{|\Omega_1|^2 + |\Omega_2|^2} \quad (12)$$

the dark and bright states are:

$$|D\rangle = \sin \theta |1\rangle - \cos \theta |2\rangle \quad (13)$$

$$|B\rangle = \cos \theta |1\rangle + \sin \theta |2\rangle \quad (14)$$

(for real $\Omega_{1,2}$). As the ratio Ω_1/Ω_2 is varied, θ rotates the dark/bright decomposition continuously through the ground-state Hilbert space.

Adiabatic Elimination of the Excited State

In the limit of large single-photon detuning $|\delta| \gg \Omega_{\text{eff}}, \gamma$, the excited state can be adiabatically eliminated ($\dot{\rho}_{ee} \approx 0$). Setting:

$$\rho_{ei} \approx \frac{\Omega_i/2}{\delta + i\gamma/2} \rho_{ii} - \frac{\Omega_j/2}{\delta + i\gamma/2} \rho_{ji} \quad (15)$$

and substituting back gives an effective **two-level Raman Hamiltonian**:

$$H_{\text{Raman}} = \hbar \begin{pmatrix} \Omega_1^2/4\delta & \Omega_1^*\Omega_2/4\delta \\ \Omega_1\Omega_2^*/4\delta & \Omega_2^2/4\delta \end{pmatrix} \quad (16)$$

The off-diagonal element $\Omega_1^*\Omega_2/4\delta$ is the **two-photon Rabi frequency**: it drives coherent transitions between $|1\rangle$ and $|2\rangle$ via virtual absorption and stimulated emission through $|e\rangle$.

Quantum Interference: Why the Dark State Cannot Absorb

The physical origin of the dark state deserves careful examination. Consider an atom in $|D\rangle$. There are two pathways by which a photon could be absorbed and the atom promoted to $|e\rangle$:

1. **Pathway 1:** Absorb a photon from beam 1 via the $|1\rangle \rightarrow |e\rangle$ transition. Amplitude:
 $\propto +\Omega_1 \cdot (\Omega_2/\Omega_{\text{eff}})$
2. **Pathway 2:** Absorb a photon from beam 2 via the $|2\rangle \rightarrow |e\rangle$ transition. Amplitude:
 $\propto -\Omega_2 \cdot (\Omega_1/\Omega_{\text{eff}})$

The total amplitude is:

$$\mathcal{A} = \frac{\Omega_1\Omega_2}{\Omega_{\text{eff}}} - \frac{\Omega_2\Omega_1}{\Omega_{\text{eff}}} = 0 \quad (17)$$

This is **destructive quantum interference between two absorption pathways**. The minus sign in $|D\rangle$ (Eq. (6)) is precisely what causes the cancellation. For the bright state $|B\rangle$, the corresponding amplitudes *add constructively*, giving the maximum coupling Ω_{eff} . This is the deepest reason why the dark/bright decomposition is the natural basis for this problem.

The two-photon resonance is essential: it ensures that both pathways share the *same final state* $|e\rangle$ with the same energy, so the interference is exact. Off two-photon resonance, the pathways lead to $|e\rangle$ with a relative phase that oscillates in time, breaking the perfect cancellation.

Coherent Population Trapping: Dynamics

Pumping into the Dark State

An atom initially in any state will be optically pumped into $|D\rangle$ by the following cycle:

1. Population in $|B\rangle$ absorbs light, is promoted to $|e\rangle$
2. $|e\rangle$ decays by spontaneous emission with rates γ_1, γ_2 back to $|1\rangle$ and $|2\rangle$
3. The new $|1\rangle, |2\rangle$ superposition is projected: the component in $|D\rangle$ is trapped; the component in $|B\rangle$ starts the cycle again
4. Eventually all population accumulates in $|D\rangle$

The **trapping time** — the average time to pump into $|D\rangle$ from an arbitrary initial state — is:

$$\tau_{\text{trap}} \sim \frac{\gamma}{\Omega_1^2 + \Omega_2^2} \quad (18)$$

This is the inverse of the optical pumping rate. For weak drives ($|\Omega| \ll \gamma$), trapping is slow; for strong drives it is fast.

Dark State Coherence

Once in $|D\rangle$, the population remains trapped as long as the **ground-state coherence** ρ_{12} is maintained. This coherence is immune to excited-state decay (since $|D\rangle$ never passes through $|e\rangle$), but is sensitive to:

- **Dephasing** of the ground states (e.g., from collisions, magnetic field noise): rate γ_{12}
- **Laser phase noise**: fluctuations in $\phi_1 - \phi_2$ rotate $|D\rangle$ into $|B\rangle$
- **Two-photon detuning errors**: $\delta_R \neq 0$ causes oscillations out of $|D\rangle$

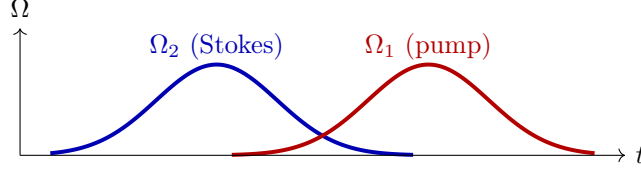
The CPT contrast (depth of the dark resonance) is determined by the ratio $\gamma_{12}/\Gamma_{\text{CPT}}$.

STIRAP: Adiabatic Transfer via the Dark State

Stimulated Raman Adiabatic Passage (STIRAP) exploits the dark state to transfer population *completely and robustly* between $|1\rangle$ and $|2\rangle$ without any population in the (lossy) excited state $|e\rangle$.

Counter-Intuitive Pulse Sequence

The key is to apply the pulses in the **counter-intuitive order**: the “Stokes” pulse (Ω_2 , coupling $|2\rangle \leftrightarrow |e\rangle$) is applied *before* the “pump” pulse (Ω_1 , coupling $|1\rangle \leftrightarrow |e\rangle$), with a temporal overlap:



Adiabatic Evolution of the Dark State

At $t \rightarrow -\infty$: $\Omega_1 = 0$, $\Omega_2 \neq 0 \Rightarrow \theta = 0 \Rightarrow |D\rangle = |1\rangle$ (the atom starts in $|1\rangle$).

At $t \rightarrow +\infty$: $\Omega_1 \neq 0$, $\Omega_2 = 0 \Rightarrow \theta = \pi/2 \Rightarrow |D\rangle = -|2\rangle$ (the atom ends in $|2\rangle$).

If the pulses are applied **adiabatically** (slowly compared to Ω_{eff}), the atom follows the instantaneous dark state $|D(\theta(t))\rangle$ throughout. The adiabaticity condition is:

$$\dot{\theta} \ll \Omega_{\text{eff}}(t) \quad (19)$$

The result is **complete population transfer** $|1\rangle \rightarrow |2\rangle$ with zero excited-state population throughout. This is robust to variations in pulse area and detuning — a key advantage over π -pulse transfer.

Electromagnetically Induced Transparency (EIT)

From CPT to EIT

EIT is the *propagation* counterpart of CPT. When a weak **probe field** (replacing Ω_1) propagates through a medium driven by a strong **coupling field** (Ω_2), the coherent dark-state preparation causes:

- **Transparency**: the absorption at the probe frequency vanishes on two-photon resonance
- **Steep dispersion**: $\partial n / \partial \omega$ is extremely large near resonance
- **Slow light**: group velocity $v_g = c / (1 + \frac{\omega}{2} \frac{\partial n}{\partial \omega})$ can be reduced to metres per second

Susceptibility and the EIT Window

The linear susceptibility seen by the probe field in the EIT medium is:

$$\chi(\omega_p) \propto \frac{\delta_R + i\gamma_{12}}{(\delta_R + i\gamma_{12})(\delta_1 + i\gamma/2) + |\Omega_2|^2/4} \quad (20)$$

At exact two-photon resonance ($\delta_R = 0$) and in the absence of ground-state dephasing ($\gamma_{12} = 0$):

$$\text{Im}[\chi(0)] = 0 \quad (\text{perfect transparency}) \quad (21)$$

The transparency window has a width $\Gamma_{\text{EIT}} \approx |\Omega_2|^2 / \gamma$, which can be made very narrow by reducing the coupling field intensity.

Dark States in Superconducting Circuits

The concept of dark states extends directly to artificial atoms (transmon qubits) coupled to microwave resonators. In a three-level transmon ($|g\rangle, |e\rangle, |f\rangle$) driven by two microwave fields:

$$|D_{\text{SC}}\rangle = \frac{\Omega_2 |g\rangle - \Omega_1 |e\rangle}{\sqrt{|\Omega_1|^2 + |\Omega_2|^2}} \quad (22)$$

This has been used experimentally for:

- **Qubit state preparation:** deterministic initialisation into a target superposition without measurement
- **Microwave EIT:** transparency windows in the transmission spectrum of the coupled qubit–resonator system
- **Single-photon generation:** using STIRAP-like protocols to transfer single excitations from a qubit to a resonator mode
- **Quantum bath engineering:** synthetic dissipation that stabilises a target dark state

Summary and Connections

Concept	Key Result
Dark state	$ D\rangle = (\Omega_2 1\rangle - \Omega_1 2\rangle)/\Omega_{\text{eff}}$; null space of coupling operator V
Bright state	$ B\rangle = (\Omega_1^* 1\rangle + \Omega_2^* 2\rangle)/\Omega_{\text{eff}}$; couples to $ e\rangle$ with Ω_{eff}
Two-photon resonance	$\omega_1 - \omega_2 = \omega_{e1} - \omega_{e2}$; required for exact CPT
CPT steady state	$\rho_{DD} = 1, \rho_{ee} = 0$; zero fluorescence
CPT linewidth	$\Gamma_{\text{CPT}} \approx \Omega_{\text{eff}}^2/\gamma \ll \gamma$
STIRAP	Counter-intuitive (Ω_2, Ω_1) order; adiabatic $ 1\rangle \rightarrow 2\rangle$ with $\rho_{ee} = 0$
Adiabaticity	$\dot{\theta} \ll \Omega_{\text{eff}}$
EIT susceptibility	$\chi(0) = 0$ at two-photon resonance; slow light $v_g \ll c$

Unifying thread: The dark state is the unique ground-state superposition whose two excitation pathways to $|e\rangle$ interfere destructively. All phenomena discussed here — CPT, STIRAP, EIT, synthetic dissipation — are consequences of this single quantum interference effect. The common thread across atomic physics, quantum optics, and superconducting circuits is the algebraic structure: the null space of the coupling operator is always non-trivial when the number of ground states exceeds the number of excited states.

Summary of Key Equations

Quantity	Expression
Λ Hamiltonian (rotating frame)	$H = \hbar \begin{pmatrix} 0 & \Omega_1^*/2 & \Omega_2^*/2 \\ \Omega_1/2 & -\delta_1 & 0 \\ \Omega_2/2 & 0 & -\delta_2 \end{pmatrix}$
Dark state	$ D\rangle = (\Omega_2 1\rangle - \Omega_1 2\rangle)/\Omega_{\text{eff}}$
Bright state	$ B\rangle = (\Omega_1^* 1\rangle + \Omega_2^* 2\rangle)/\Omega_{\text{eff}}$
Effective Rabi frequency	$\Omega_{\text{eff}} = \sqrt{ \Omega_1 ^2 + \Omega_2 ^2}$
Two-photon resonance	$\delta_1 = \delta_2$, i.e. $\omega_1 - \omega_2 = \omega_{e1} - \omega_{e2}$
CPT linewidth	$\Gamma_{\text{CPT}} \approx \Omega_{\text{eff}}^2/\gamma$
Raman Hamiltonian	$H_{\text{Raman},12} = \hbar \Omega_1^* \Omega_2 / 4\delta$
STIRAP adiabaticity	$\dot{\theta} \ll \Omega_{\text{eff}}(t)$
EIT susceptibility	$\chi \propto (\delta_R + i\gamma_{12})/[(\delta_R + i\gamma_{12})(\delta_1 + i\gamma/2) + \Omega_2 ^2/4]$